BTC funding carry: P&L; report

Historical simulation | 10,000 USDT initial capital | prepared 16 October 2026

A fully funded BTC spot holding is hedged by an equal-quantity short linear perpetual. Funding is collected or paid while positions remain open. Monthly rebalances trade only the change in BTC quantity. Every test ends with both positions closed. This is a price-proxy backtest, not live or order-book-verified performance.

Account-level profit and loss

USDT unless stated	2022-2024 development	Feb-Dec 2025 later test
Initial capital	10,000.00	10,000.00
Funding income	1,356.32	223.62
Basis P&L	-6.00	-0.22
Trading fees	-53.06	-21.86
Assumed slippage	-14.15	-5.83
Net P&L	1,283.11	195.70
Final cash, positions closed	11,283.11	10,195.70
Total return	12.83%	1.96%
Maximum daily drawdown	-0.18%	-0.09%

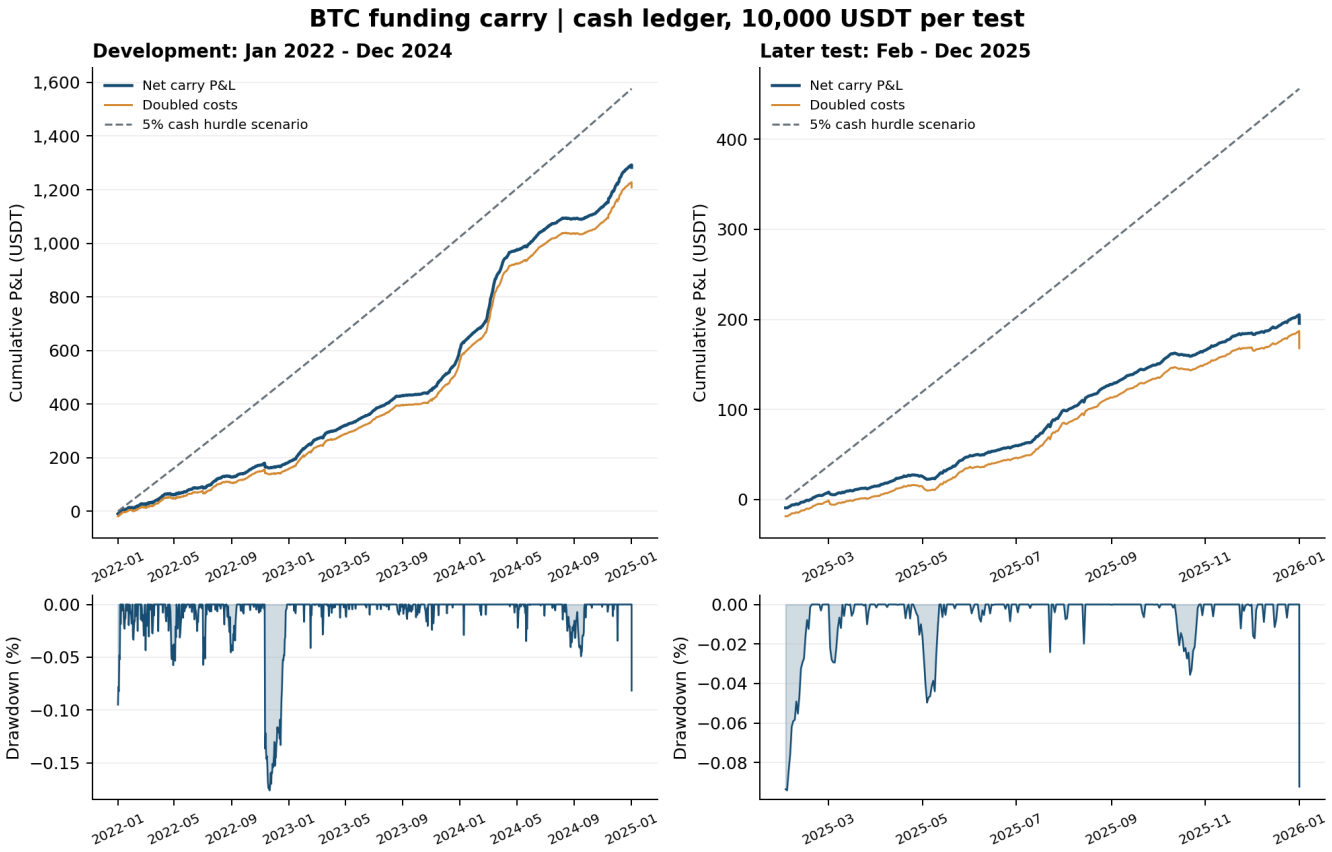
Calendar-year attribution

Returns below use the running account balance. The later test starts from a separate 10,000 USDT account and covers 11 months. Terminal January 1 funding and final close costs are included in the concluding December period.

Period	Net P&L (USDT)	Return on capital
2022	184.26	1.84%
2023	418.98	4.11%
2024	679.88	6.41%
2025 (Feb-Dec)	195.70	1.96%

Positive nominal P&L; is not sufficient evidence of an attractive investment. Compare each test with the capital hurdle and account for unmodeled venue and stablecoin risks. No predictive entry signal has been fitted.

Equity, drawdown and cost stress



Scenario / comparison	Development	Later test
Base-cost net P&L	1,283.11	195.70
Doubled-cost net P&L	1,208.30	168.01
5% cash hurdle: terminal value	11,576.64	10,456.26
Strategy minus cash hurdle	-293.53	-260.56
Worst valuation-interval P&L	-13.89	-9.40
Minimum futures cash balance	1,317.55	4,090.37
Minimum stressed margin buffer	2,632.31	4,045.09

The cash hurdle is an illustrative 5% annual compounding scenario with the same starting capital and elapsed time. It is not an audited historical investment or an assumed return on collateral. The strategy earns no collateral interest in this simulation.

Drawdown is measured at observed midnight valuations plus entry and final-close endpoints. Small measured drawdowns do not include exchange failure, stablecoin depegging, liquidation gaps or inability to trade during outages. The chart is not a forecast.

Ledger, validation and limits

Cash and valuation identity

Equity = USDT cash + BTC quantity x spot price + short BTC quantity x (average perpetual entry price - mark price). The ledger tracks perpetual entry averages, partial-close realized P&L;, funding cash flows and each cost debit. Final positions are zero, so terminal equity is cash.

P&L; = funding + spot price P&L; + perpetual price P&L; + the execution-price/mark adjustment - spot fees - perpetual fees - slippage. Every hourly observation reconciles this attribution to account equity within 0.000001 USDT.

Trading and risk assumptions

At 01:00 UTC on each first day of the month, target roughly half the account in BTC spot and retain the rest as futures collateral. Both legs have the same quantity, rounded down to a 0.001 BTC research lot. This lot is a modeling restriction, not a reconstruction of every historical exchange order filter. Spot fee: 10 basis points per execution. Perpetual fee: 5 basis points. Additional slippage: 2 basis points per leg per execution. Stress doubles all four cost coefficients.

All USDT not invested in spot is assigned to futures collateral during holding periods. Monthly spot funding uses actual available cash; no borrowing against unrealized profit is required. Initial margin is checked against a 50% notional scenario. Maintenance stress uses 1% of notional at the hourly mark high, debits negative funding within the hour and does not credit positive funding early. Any breach causes the replay to fail rather than report a completed profit.

Historical validation and checks

The 2022-2024 fractional-quantity cash ledger reproduces the earlier terminal result within 0.000001 USDT. The primary report adds lot rounding. Strategy, sizing, cost and margin rules were recorded before downloading February 2025 onward data. January 2025 was already downloaded and is excluded from the later test. A report-only annual grouping fix is documented; no positions or strategy rules changed after freezing.

Eight accounting tests pass, covering directional cancellation, funding booking time, partial-close realization, lot sizing, fee debits, collateral allocation and avoiding redundant round trips. These verify accounting, not profitability. Later-period performance is a single historical validation, not a broad statistical proof.

Data and practical limitations

Official Binance monthly spot, perpetual, mark-price and funding archives are used. Source URLs and hashes are retained. Three missing development mark days were restored from official daily archives. One development spot hour has a stale valuation, explicitly flagged; no entry, exit or daily report mark uses a stale price. Funding uses native settlement timestamps and scheduled-hour mark opens as settlement-price proxies. Execution uses hourly trade-price opens plus assumed costs.

Bid/ask depth, legging risk, actual historical fee tiers and margin rules, sub-hour liquidation paths, operational transfers, venue solvency, stablecoin dollar value and taxes are not reconstructed. Archive event time and completeness do not prove historical data publication availability. Continuous equal-quantity hedging removes the main BTC direction exposure but leaves funding and basis risk.

Reproduce and inspect

See daily_statement_1x.csv for interval-level P&L; trades_1x.csv for execution and realized futures P&L; funding_1x.csv for every payment; hourly_1x.csv.gz for marked positions, cash and margin checks. Run pnl.py for development and pnl.py --validation for the later test. PNL_PROTOCOL.json records frozen choices. Source archives and manifests are included.

Primary references: github.com/binance/binance-public-data; binance.com/en/support/faq/detail/360033525031. Economic background: Schmeling, Schrimpf and Todorov, Crypto Carry, BIS Working Paper 1087 (2023).

Performance KPIs: what to report

Sharpe must identify its cash benchmark. A ratio calculated against idle cash describes return per unit of observed volatility, but does not establish value above the opportunity cost of capital. Daily volatility also excludes rare venue and stablecoin loss events.

Carry strategy: matched 24-hour observations

Metric	2022-2024	Feb-Dec 2025
Total net return	12.83%	1.96%
Annualized return (CAGR)	4.11%	2.14%
Volatility from daily returns	0.36%	0.22%
Volatility from hourly returns	1.21%	0.62%
Sharpe above SOFR, daily returns	0.09	-9.68
Sharpe above SOFR, hourly returns	0.03	-3.53
Sharpe above zero, hourly returns	3.32	3.44
Maximum hourly drawdown	-1.12%	-0.11%
Daily return observations	1096	334

Metrics use 01:00 UTC marks exactly 24 hours apart, with entry costs in the first daily return and final exit costs in the last. This differs from the midnight sampling on earlier P&L; pages; final cash is unchanged. Maximum hourly drawdown is the more conservative observed drawdown measure. One development spot hour is flagged stale; the maximum hourly drawdown occurs at a different, fresh observation.

Definitions

Sharpe = sqrt(365.25) x mean(daily strategy return - daily cash return) / standard deviation(daily excess return). Annual volatility = daily strategy-return standard deviation x sqrt(365.25). CAGR compounds terminal wealth over elapsed years; it is not an arithmetic mean. The 11-month annualized figure is descriptive and is not an extra year of observations.

Hourly volatility and ratios use sqrt(365.25 x 24) annualization. The daily zero-cash ratios of 11.11 and 9.43 fall to 3.32 and 3.44 with hourly returns. Thus the largest number is highly sampling-dependent. Serial dependence also affects annualization; HAC and block-bootstrap inference below account for dependence when testing means. No simple frequency-scaled ratio fully captures venue tail risk.

Historical cash benchmark

Historical SOFR rates come directly from the New York Fed. The cash proxy retrospectively accrues each effective-day rate on ACT/360, carrying the previous business-day rate across weekends and holidays. It is an accrual proxy, not the official SOFR Index, an investable deposit account, or guaranteed USDT income. USD/USDT parity is assumed for comparison. The strategy itself receives no added collateral interest. SOFR is not used by any trading signal.

Benchmarks and statistical evidence

Net cumulative returns over matching dates

Benchmark	2022-2024	Feb-Dec 2025
Continuous funding carry	12.83%	1.96%
BTC buy-and-hold	101.81%	-14.47%
20-day momentum, long / cash	10.72%	-20.73%
Historical SOFR cash proxy	12.72%	4.01%

Spot benchmarks use the same 10bp fee, 2bp slippage and 0.001 BTC research lot. Buy-and-hold invests once and closes at the end. Momentum holds BTC when the price change across 20 calendar days is positive; its signal uses the prior day 00:00 price and executes at 01:00 the following day, at least 25 hours later. Cash while flat earns zero. No parameter search is used. These are descriptive benchmarks, not newly validated optimal strategies.

Risk differs materially: buy-and-hold maximum daily drawdown was 66.66% in development and 32.03% in the later sample, compared with carry hourly drawdowns of 1.12% and 0.11%. Higher BTC returns in one sample are not a comparison at equal risk. Carry beat the two spot benchmarks in the later realized sample, but neither paired daily-mean comparison establishes statistically reliable outperformance.

Is excess return statistically established?

Carry minus historical SOFR	2022-2024	Feb-Dec 2025
Annualized mean daily excess	0.03%	-2.18%
95% block-bootstrap interval	[-1.06%, 1.26%]	[-2.81%, -1.50%]
HAC two-sided p-value	0.955	2.05e-11

Intervals use 10,000 circular moving-block bootstrap draws, 30-day blocks and seed 16. HAC tests use 30 lags. The cash-excess conclusion is unchanged with 7-day and 90-day choices: development intervals include zero; the later intervals are wholly negative. Intervals concern annualized arithmetic mean daily excess returns, not a confidence interval for the realized cumulative return or CAGR.

Within this backtest, positive returns above idle cash are supported, but development excess return above SOFR is not. The later sample shows underperformance against the cash proxy under the dependence-aware tests. These are exploratory conditional results. Stationarity, bootstrap and covariance assumptions can fail across regimes; limited tail events and price/execution errors are not repaired by a low p-value. Paired benchmark comparisons are not adjusted for multiple testing, and no broad alpha claim is made.

Use on a CV

Lead with the self-financing ledger, funding and margin modeling, frozen later-period evaluation and cost stress. Quantify 1.96% net over the 11-month later sample, 0.11% maximum observed hourly drawdown and 168.01 USDT profit under doubled costs on 10,000 USDT. Label all figures as simulated. Do not advertise a Sharpe of 9.43 without its zero-cash benchmark; the cash-excess Sharpe is negative. This remains a research/backtesting project with no demonstrated premium over cash.

Sources: New York Fed SOFR; Lo (2002), Financial Analysts Journal; Newey-West (1987), Econometrica. Eleven checks pass. Reproduce with kpis.py; full results are in results/kpis/.